Addis Ababa Science and Technology University

College of Natural and Applied Science

Department of MPS

General Physics (Phys 1001) worksheet I

Group assignment 1(a group of 5 members): (#8, #9, #11, #12, #13, #14, #15, #19, #21, #22)

Submission Date: _____

1. If the component vector $\vec{A}$ along the direction of vector $\vec{B}$ is zero, what can you conclude about the vectors?

2. a) Is it possible to add a vector quantity to a scalar quantity? Explain
b) Can the magnitude of a vector have a negative value? Explain

3. Which of the following are vectors and which are not: force, temperature, the volume of water in a can, the ratings of a TV show, the height of a building, the velocity of a sports car, the age of the Universe?

4. Determine the number of significant figures for the following numbers:

90000	70000000.0	84.10000
10082	0.0025	3008000
70000000	0.00008914	0.000339

5. A jogger runs 145 m in a direction 20.0° east of north and then 105 m in a direction 35.0° south of east. Using components, determine the magnitude and direction of the resultant vector for these two displacements.

6. A particle undergoes three consecutive displacements: $\vec{d}_1 = (15i + 30j + 12k)cm$, $\vec{d}_2 = (23i - 14j - 5k)cm$ and $\vec{d}_3 = (-13i + 15j)cm$ Find the magnitude of the resultant displacement.

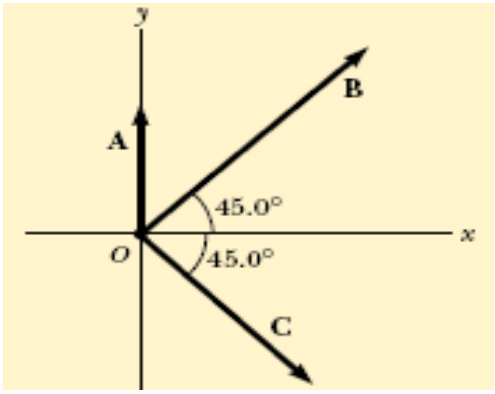
7. Consider the two vectors $\vec{A} = 3i - 2j$ and $\vec{B} = -i - 4j$ then Calculate:

(a) $\vec{A} + \vec{B}$ (b) $\vec{A} - \vec{B}$ (c) $|\vec{A} + \vec{B}|$ (d) $|\vec{A} - \vec{B}|$

8. Given the two vectors $\vec{A} = 2i + j - k$ and $\vec{B} = -3i + 2j + k$ then find:

(a) a unit vector in the direction of $-3\vec{A} + 2\vec{B}$ (b) $\vec{A} \bullet \vec{B}$ (c) $(\vec{A} + \vec{B}) \bullet (\vec{A} - \vec{B})$

9. Three displacement vectors of a cricket ball are shown in the Figure below, where $|\vec{A}| = 20$ units, $|\vec{B}| = 40$ units, and $|\vec{C}| = 30$ units. Find the magnitude and direction of the resultant displacement.



10. Vector $\vec{A}$ lies in the $y-z$ plane of 63^0 from the positive $y-axis$ and has magnitude of 3.2 $unit$ Vector $\vec{B}$ lies in the $x-z$ plane of 48^0 from the positive $x-axis$ and has magnitude of 1.4 $unit$ Find the $\vec{A}.\vec{B}$, $\vec{A}x\vec{B}$ and the angle between $\vec{A}$ and $\vec{B}$

11. Given three vectors $\vec{A}=(2i-3j+4k)m$ and $\vec{B}=(-3i-j+2k)m$ and $\vec{C}=(2i+2j+k)m$ then find:

(a) The angle between vectors $\vec{A}$ and $\vec{B}$

(b) Unit vector along the direction of $\vec{B}$.

(c) Unit vector perpendicular to both vectors $\vec{B}$ and $\vec{C}$.

(d) Area of the triangle with $\vec{A}$ and $\vec{B}$ as its sides.

(e) Area of the parallelogram with $\vec{B}$ and $\vec{C}$ as its sides.

12. Vector $\vec{A}$ has x and y components of -8.7cm and 15cm respectively; vector $\vec{B}$ has x and y components of 13.2cm and -6.6cm respectively. If $\vec{A}-\vec{B}+3\vec{C}=0$, What are the x and y components of vector $\vec{C}$?

13. Given a unit vector $\hat{u}=\frac{a}{c}\hat{i}+\frac{2b}{c}\hat{j}$ where a, b and c are positive scalars. Find c and express its value in terms of a and b.

14. A certain force vector is expressible as $\vec{F}=(10i+bj)N$. What is the value of the constant b if the magnitude of the force vector is 26N?

15. Let $|\vec{A}|=150N$ and the vector $\vec{A}$ makes an angle of 60^0 with the positive y axis. Let P be the scalar projection of $\vec{A}$ on to the xz plane and it makes an angle of 30^0 with the positive x axis. Express vector $\vec{A}$ in terms of its rectangular(x,y,z) components;

16. For what value of a are the vectors $\vec{A}=2i+4j+6k$ and $\vec{B}=3i+3a-3k$ perpendicular to each other?

17. If $\vec{A}$ and $\vec{B}$ are unknown vectors such that $\vec{A}+3\vec{B}=-3i+j$ and $\vec{A}-\vec{B}=i-2j$,then find the expression for the vectors $\vec{A}$ and $\vec{B}$?

18. Let vectors $\vec{A} = 2i - 3j$ and an unknown vector $\vec{B}$ lie on the xy plane, where $\vec{A} \bullet \vec{B} = 4$ and $\left|\vec{A} \times \vec{B}\right| = 4$

A) Find the angle θ between the two vectors

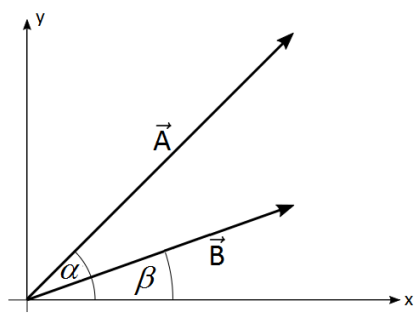
B) Find $\vec{B}$ and express it in terms of rectangular unit vectors

19. Consider vector $\vec{A} = ni + mj$ and vector $\vec{B}$ in which n and m are scalars. If $\vec{A} \bullet \vec{B} = 2nm$ and $\vec{A} \times \vec{B} = (n^2 - m^2)k$ then find $\vec{B}$ and express it in terms of n, m

20. If $\vec{A} = 2\hat{\imath} - \hat{\jmath} + \hat{k}$ and $\vec{B} = \hat{\imath} + 2\hat{\jmath} - 3\hat{k}$ then evaluate $\left(2\vec{A} + \vec{B}\right)x\left(\vec{A} - 2\vec{B}\right)$

21. In the diagram below, $\vec{A}$ is a vector of magnitude 35 cm; $\vec{B}$ is a vector of magnitude 13 cm. If

Tanα = 4/3 and tanβ = 5/12, write A and B in terms of $\hat{\imath}$ and j



22. For two vectors $\vec{A}$ and $\vec{B}$, if $\vec{A}$ x $\vec{B}$ = $6\,\hat{\imath} + 2\hat{\jmath} + 5\hat{k}$, $\vec{A} \cdot \vec{B} = -13$, $\vec{A}$ + $\vec{B}$ = $-2\hat{\imath} + \hat{\jmath} + 2\hat{k}$ and $\left|\vec{A}\right| = 3$, then find:

a) The two vectors $\vec{A}$ and $\vec{B}$

b) The area of the parallelogram and the triangle having $\vec{A}$ and $\vec{B}$ as their edge.

c) A third vector having magnitude twice $\vec{A}$ and direction perpendicular to both vectors.